

GEOMETRY OF GAUGE FIELDS

(in progress)

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ABSTRACT. Gauge theory is the unifying framework underlying the fundamental interactions of nature, including electromagnetism, the weak, the strong, and more subtly, the gravitation. Their formulation is inherently mathematical relying on the theory of connections on fibre bundles and the associated curvature. In this exposition, we present these geometric foundations of gauge theory and relate them to the physical theories of electromagnetism, Yang-Mills, and general relativity. We also outline their influence on low-dimensional topology, particularly through their connections to Knot theory and four-manifold invariants.

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1. LIE THEORY

1.1. Lie Groups.

Definition 1.1. A **Lie group** is a group G which is also a smooth manifold with smooth group operations,

$$\begin{aligned}\cdot : G \times G &\rightarrow G \\ ^{-1} : G &\rightarrow G.\end{aligned}$$

Behind all the Yang-Mills equations, there is a lingering Lie group structure that induces the whole physics. The Lie Groups which occur as the symmetry groups in physics, can themselves be studied with the tools of smooth calculus. This ties intimately to the tangent space at the identity through which the whole G can be generated using the calculus on manifolds.

The most common examples which appear in physics include, the general linear groups $GL(n, \mathbb{R})$ and $GL(n, \mathbb{C})$, the special linear groups $SL(n, \mathbb{R})$ and $SL(n, \mathbb{C})$,

the orthogonal group $O(n)$, the special orthogonal group $SO(n)$, the lorentz group $SO(p, q)$ and $O(p, q)$, and the unitary groups $U(n)$ and $SU(n)$. The manifold structure of these groups follows from the *Regular value theorem on manifolds*, using maps like $F : M_n(\mathbb{R}) \rightarrow SM_n(\mathbb{R}), F(A) = AA^T$ or $\det : M_n(\mathbb{R}) \rightarrow \mathbb{R}!$

Definition 1.2. Given a Lie Group G . The identity component G^+ of G is the connected component containing the identity element of G .

Proposition 1.3. *The identity component G^+ of a Lie Group G is a subgroup and a Lie Group in its own right.*

Proof. The connected component of a manifold must be open, and is thus a sub-manifold, with the group operations $\cdot$ and $^{-1}$ smooth. G^+ is further closed under invverses because the map $^{-1} : G \rightarrow G$ is smooth and G^+ has to be mapped to G^+ as a connected component. Similarly we see that $G^+ \times G^+ \mapsto G^+$. Thus, $G^+ \leq G$. $\square$

Some familiar examples from a course in relativity or quantum field theory are as follows.

Example 1.4. The identity component of $O(3)$ is $SO(3)$, and any element of $O(3)$ can be written as PR , where $R \in SO(3)$, rotation around some axis and P is the reflection through some plane.

Example 1.5. $SO(3, 1)$ has two connected components. The identity element and PT (partity $\times$ time-reversal) are not in the same component. This is also called, the proper orthochronus Lorentz group, which preserves both the time direction and the spatial orientation! The group $O(3, 1)$ in general splits into four connected components which as a set of four elements has the group structure, $O(3, 1)/SO^+(3, 1)$, precisely corresponding to the cosets given by $\{I, P, T, PT\}$.

It becomes important to study various ways in which these groups can act (on say vector spaces).

Definition 1.6. A group G acts on a vector space V if there is a map ρ from G to the linear transformations of V , $\rho : G \rightarrow GL(V)$ such that,

$$\rho(gh)v = \rho(g)\rho(h)v.$$

We call ρ to be a **representation** of G on V . This is nothing but a Lie Group homomorphism between G and $GL(V)$.

In particular, it will be very useful to classify representations of groups for which we need the notion of two representation being equal.

Definition 1.7. Two representations $\rho : G \rightarrow GL(V)$ and $\rho' : G \rightarrow GL(V')$ are said to be equivalent, if there is a linear isomorphism $T : V \rightarrow V'$ such that,

$$\rho'(g)T = T\rho(g) \quad \forall g \in G.$$

The basic representations one can immediately write off are:

- (1) *Fundamental representation.* Matrix groups already lie in $GL(V)$, with $V = \mathbb{R}^n$ or $\mathbb{C}^n$. So, by definition they can act on V !
- (2) *Direct sum representation.* Given representations ρ, ρ' on V, V' , we can define a representation on $V \oplus V'$ by

$$(\rho \oplus \rho')(g)(v, v') = (\rho(g)v, \rho'(g)v').$$

- (3) *Tensor product representation.* Given two representations ρ, ρ' on V, V' , we can define a representation on $V \otimes V'$ by

$$(\rho \otimes \rho')(g)(v \otimes v') = \rho(g)v \otimes \rho'(g)v'.$$

We can also look *into* a representation and obtain a subrepresentation.

Definition 1.8. Given a representation ρ of G on V , suppose that V' is an invariant subspace of V , i.e. $\rho(g)v \in V'$ for all $v \in V', g \in G$. Then, we can define a **subrepresentation** ρ' of G on V' by setting

$$\rho'(g)v = \rho(g)v.$$

If there are no invariant subspaces of V other than $\{0\}$ and V , then ρ is said to be **irreducible**.

These irreducible representations occur as building blocks for more general representations.

Theorem 1.9 (Complete reducibility of Compact groups). *If G is compact, then every representation is equivalent to a direct sum of irreducible representations.*

All the gauge groups that occur in physics, except general relativity are indeed compact. And it is these irreducible representations that describe the elementary particles in Quantum field theory.

1.2. Irreps of $U(1)$. For reasons at the heart of gauge theory, $U(1)$ is considered to be the gauge group of electromagnetism. Finding its irreducible representations offers an immediate way connect to matters of charge quantization and magnetic monopoles. We will precisely do that in this section. This will be ultimately connected the non-trivial topology of $U(1)$ bundle in later sections.

$U(1) = \{e^{i\theta} \mid \theta \in \mathbb{R}\}$ is just the unit circle in complex plane. It has representations ρ_n on $\mathbb{C}$,

$$\begin{aligned} \rho : U(1) &\rightarrow GL(\mathbb{C}) \\ \rho_n(e^{i\theta})v &= e^{in\theta}v. \end{aligned}$$

These are clearly irreducible since, $\mathbb{C}$ as a vector space over itself, has no other subspaces other than $\{0\}$ and the whole space. Infact, the other direction is also true.

Proposition 1.10. *Any complex irreducible representation of $U(1)$ is equivalent to one of the representations ρ_n .*

Proof. Consider any complex 1-dimensional representation, $\rho' : U(1) \rightarrow GL(\mathbb{C})$. We can first show that this is equivalent to some ρ_n . $\square$

$U(1)$ is also compact so all finite-dimensional representations can be built using these irreps ρ_n . This naturally encompasses the gauge symmetry implications in quantum electromagnetism. When a charged particle of charge nq is moved around a loop in spacetimes, its wavefunction admits a phase change, which can be seen quite easily using the path integral,

$$\psi \rightarrow e^{inq \oint_\gamma A} \psi$$

where A is the vector potential. In the units of $\hbar = q = 1$, this is just,

$$\rho_n \left(e^{-i \oint_\gamma A} \right)$$

where $e^{-i \oint_\gamma A}$ lives on $U(1)$ reflecting the fact that the gauge group representations dictate the phase change of the wavefunctions when a particle is spun around a loop. And further, different representations of $U(1)$ dictate different possible charged particles, to be integer multiples of some basic charge q . This way $U(1)$ as a gauge group makes magnetic monopoles natural, since one of their requirements is charge quantization. Further, the tensor product representation due to ρ_n and ρ_m corresponds to the representation ρ_{n+m} . This manifests physically as the additivity of total charge of two particles in a bound state. The goal is to generalize this picture to higher groups like $SU(2)$ eventually leading us to the Yang-Mills equations.

1.3. Irreps of $SU(2)$. We know that $SU(2)$ is spanned by the Pauli matrices along with the identity, and the algebra of Pauli matrices is precisely that of quaternions.

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

And $SU(2)$ is precisely the 3-sphere in the quaternions, like $U(1)$ was a 1-sphere in complex plane.

$$SU(2) = \{ aI + b\sigma_x + c\sigma_y + d\sigma_z \mid a, b, c, d \in \mathbb{R} \text{ } a^2 + b^2 + c^2 + d^2 = 1 \}$$

The irreducible representations of $SU(2)$ are the famous spin- j representations, U_j exactly one (upto equivalence) for each dimension $2j + 1$ given by,

$$U_j : SU(2) \rightarrow GL(\mathcal{H}_j) \\ g \mapsto (U_j(g)f)(v) := f(g^{-1}v).$$

Here $\mathcal{H}_j$ is the space of polynomial functions on $\mathbb{C}^2$ that are homogenous of degree $2j$. That is any $h \in \mathcal{H}_j$ is a linear combination of the polynomials,

$$f(x, y) = x^p y^q$$

where $p + q = 2j$. Thus $\mathcal{H}_j$ has a dimension $2j + 1$. As can be checked easily, the inverse is indeed required for U_j to be a representation.

Proposition 1.11. *The spin-0 and spin-1/2 representations of $SU(2)$ are equivalent to the trivial and fundamental representations respectively.*

Proof. The spin-0 representation is over $\mathcal{H}_0$ which is just the constant polynomial. And this is trivially isomorphic to $\mathbb{C}$. With this, the following diagram commutes

$$\begin{array}{ccc} \mathcal{H}_0 & \xrightarrow{U_0} & \mathcal{H}_0 \\ T \downarrow & & \downarrow T \\ \mathbb{C} & \xrightarrow{\rho} & \mathbb{C} \end{array}$$

since if $f \in \mathcal{H}_0$ and $f(x) = w_0 \in \mathbb{C}$,

$$\begin{aligned} T(U_0(g)(f(x))) &= T f(g^{-1}x) \\ &= T f(x) \\ &= w_0 \\ &= \rho(g) T f(x) \end{aligned}$$

where $\rho(g)(z) = z \forall g \in SU(2)$ is the trivial representation.

The spin-1/2 representation is over $\mathcal{H}_{\frac{1}{2}}$, whose polynomials are simply, $f(x, y) = ax + by$. We will write this in the form $f(\psi) = \phi^T \psi$ for some $\phi \in \mathbb{C}^2$, where we also denoted $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{C}^2$ by ψ . We can write $U_{\frac{1}{2}}$ in this notation as,

$$U_{\frac{1}{2}} f(\psi) = f(g^{-1}\psi)$$

Now consider the map,

$$\begin{aligned} T : \mathbb{C}^2 &\rightarrow \mathcal{H}_{\frac{1}{2}} \\ \varphi &\mapsto \varphi^T \psi \end{aligned}$$

where $\epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ satisfying $g^T \epsilon g = \epsilon \implies g^T \epsilon = \epsilon g^{-1}$. Thus,

$$\begin{aligned} U_{\frac{1}{2}}(g)T\varphi &= \varphi^T \epsilon g^{-1} \psi \\ &= \varphi^T g^T \epsilon \psi \\ &= (g\varphi)^T \epsilon \psi \\ &= T(g\varphi) \\ &= T\rho(g)\varphi \end{aligned}$$

where $\rho(g) = g$ is the fundamental representation. Thus, the following diagram commutes.

$$\begin{array}{ccc} \mathcal{H}_{\frac{1}{2}} & \xrightarrow{U_{\frac{1}{2}}} & \mathcal{H}_{\frac{1}{2}} \\ \uparrow T & & \uparrow T \\ \mathbb{C}^2 & \xrightarrow{\rho} & \mathbb{C}^2 \end{array}$$

□

The importance of these irreps U_j however comes especially from the spin-1 representation, which is related to the group of rotations in 3-dimensions, $SO(3)$.

Proposition 1.12. *There is a Lie Group homomorphism*

$$\rho : SU(2) \rightarrow SO(3).$$

This can be seen as a 3-dimensional complex representation of $SU(2)$ on $\mathbb{C}^3$, since $SO(3) \subset GL(3, \mathbb{R}) \subset GL(3, \mathbb{C})$ as a Lie subgroup.

Proof. First note that the traceless hermitian matrices which are spanned by Pauli-matrices with real coefficients, are invariant under the action of $SU(2)$:

$$\rho(g)T = gTg^{-1}$$

$$\text{tr}(gTg^{-1}) = \text{tr}T = 0$$

$$\begin{aligned} (gTg^{-1})^\dagger &= (g^{-1})^\dagger T^\dagger g^\dagger \\ &= gTg^{-1}. \end{aligned}$$

The space of traceless hermitian matrices can be identified with $\mathbb{R}^3$ using the basis of Pauli matrices and we thus have,

$$\rho : SU(2) \rightarrow GL(3, \mathbb{R}).$$

Let $(T_1, T_2, T_3) \in \mathbb{R}^3$, it's length is precisely the determinant of its correspond traceless hermitian matrix, and

$$\det(\rho(g)T) = \det(gTg^{-1}) = \det(T).$$

So, the lengths in $\mathbb{R}^3$ are preserved by $\rho(g)$ for all $g \in SU(2)$. Thus ρ maps further to $O(3) \leq GL(3, \mathbb{R})$. Infact, since $SU(2)$ is connected, ρ maps it to the identity component of $O(3)$ which is $SO(3)$! $\square$

Proposition 1.13. *The above homomorphism $\rho : SU(2) \rightarrow SO(3)$ is two-to-one.*

Proof. Clearly, $\rho(g) = \rho(-g)$. And let's assume this to be an irreducible representation of $SU(2)$. Suppose $\rho(g) = \rho(h)$, then:

$$\rho(gh^{-1}) = \rho(g)\rho(h)^{-1} = 1.$$

This means, gh^{-1} commutes with all the traceless hermitian matrices! But by Schur's lemma, $gh^{-1} = \lambda I$, and since $gh^{-1} \in SU(2)$ so $\lambda = \pm 1$. Thus, $g = \pm h$. We will show that the representation ρ is indeed irreducible by showing its equivalence with spin-1 representation! However, we omit the proof for the irreducibility of spin-j representation. $\square$

Proposition 1.14. *The double cover $\rho : SU(2) \rightarrow SO(3)$ is equivalent to the spin-1 representation of $SU(2)$, $U_1 : SU(2) \rightarrow GL(\mathcal{H}_1)$.*

Proof. Let's consider the representation $\rho : SU(2) \rightarrow SO(3)$ to be over the space of complex traceless matrices $\mathcal{V}_2$, i.e.

$$\rho : SU(2) \rightarrow GL(\mathcal{V}),$$

Our goal is to give a linear isomorphism T between $\mathcal{H}_1$ and $\mathcal{V}$ such that the following diagram commutes.

$$\begin{array}{ccc} \mathcal{H}_1 & \xrightarrow{U_1} & \mathcal{H}_1 \\ \uparrow T & & \uparrow T \\ \mathcal{V} & \xrightarrow{\rho} & \mathcal{V} \end{array}$$

Any polynomial in $\mathcal{H}_1$ is of the form $ax^2 + bxy + cy^2$. Define the polynomial associated with the matrix $M \in \mathcal{V}$ to be,

$$T(M)(\psi) = \psi^T \epsilon^T M \psi, \quad \psi \in \mathbb{C}^2$$

with the same ϵ defined in the case of spin-1/2 representation. We can expand this to see the isomorphism,

$$\begin{pmatrix} -\frac{b}{2} & -c \\ a & \frac{b}{2} \end{pmatrix} \mapsto ax^2 + bxy + cy^2.$$

The equivalence condition can be checked:

$$\begin{aligned}
U_1(g)T(M) &= (g^{-1}\psi)^T \epsilon^T M g^{-1}\psi \\
&= \psi^T (g^{-1})^T \epsilon^T M g^{-1}\psi \\
&= \psi^T \epsilon^T g M g^{-1}\psi \\
&= \psi^T \epsilon^T (g M g^{-1}) \psi \\
&= T\rho(g)M.
\end{aligned}$$

□

Proposition 1.15. *The above homomorphism $\rho : SU(2) \rightarrow SO(3)$ is onto.*

Proof. We defer this proof to Lie Algebras. □

This Lie algebra homomorphism $\rho : SU(2) \rightarrow SO(3)$ is called the **double cover** of $SO(3)$. And occurs as an important tool to study the group of rotations $SO(3)$ itself. In the follow section we see how this double cover gives a complete list of (projective) irreducible unitary representations of $SO(3)$. It also distinguishes the half-integer and integer spin representations viz. the bosonic and fermionic cases. This distinction comes from the flexibility in quantum mechanics to work with projective unitary representations.

From the lens of transition probabilities, there are equivalent classes of states in a Hilbert space,

$$[\psi] = \{e^{i\theta}\psi : \theta \in \mathbb{R}\}$$

related mutually by a simple phase factor. This allows for an expanded collection of unitary representations of a group G acting on a complex vector space.

Definition 1.16. A representation, $\rho : G \rightarrow GL(\mathcal{H})$ is called **projective** if it satisfies the modified representation rules,

$$\rho(1) = e^{i\theta}, \quad \rho(g)\rho(h) = e^{i\theta(g,h)}\rho(gh)$$

where θ is a fixed real number and $e^{i\theta(g,h)}$ is called a cocycle, function of $g, h \in G$.

Remark 1.17. In some cases it is possible to get rid of this cocycle, by defining another representation $\rho'(g) = e^{i\varphi(g)}\rho(g)$ which could work as an honest representation. If no such choice of $\varphi(g)$ exists, then the associated residual cocycle is called **essential**. This marks a fundamental difference in bosons and fermions when realized as irreducible representations of the rotation group $SO(3)$.

1.4. Irreps of $SO(3)$. The polynomials in $\mathcal{H}_j$ which is the spin- j representation space of $SU(2)$, has a degree $2j$. So for any $f \in \mathcal{H}_j$

$$(U_j(-1)f)v = f(-v) = (-1)^{2j}f(v).$$

1.4.1. *Bosonic Case.* Recall that as a spin-1 representation the double cover $\rho : SU(2) \rightarrow SO(3)$ maps -1 and 1 to the identity of $SO(3)$. We now see that, for any integer j values, U_j maps both -1 and 1 to the identity. So,

$$U_j(g) = U_j(-g)$$

for all $g \in SU(2)$. But since $\rho(-1) = 1$, we have $\rho(g) = \rho(-g)$ for all $g \in SU(2)$.

For any $h \in SO(3)$ we can choose either g , or $-g \in SU(2)$ such that $\rho(g) = \rho(-g) = h$. We can then construct a representation of $SO(3)$,

$$\begin{aligned} V_j : SO(3) &\rightarrow GL(\mathcal{H}_j) \\ h &\mapsto V_j(h) = U_j(g) \end{aligned}$$

which is well-defined for integral j values. Indeed,

$$V_j(hh') = U_j(gg') = U_j(g)U_j(g') = V_j(h)V_j(h')$$

and $V_j(1) = U_j(1) = U_j(-1) = 1$. Thus the double cover ρ gives integer spin irreducible representations of $SO(3)$.

1.4.2. *Fermionic Case.* Now consider the half-integer j values, for which $U_j(-1) = -1, U_j(1) = 1$. We can proceed with the same choice of representation and reach the ambiguity of choosing $g, -g \in SU(2)$ so that $\rho(g) = \rho(-g) = h \in SO(3)$ but

$$U_j(g) = -U_j(-g) = 1.$$

With this, for any $h, h' \in SO(3)$, we have by some choice $g, g' \in SU(2)$ and

$$V_j(hh') = U_j(\pm gg') = \pm U_j(g)U_j(g') = \pm V_j(g)V_j(h').$$

This cocycle turns out to be essential. So, the half-integer spin representations of $SU(2)$ give atmost a projective unitary representations of $SO(3)$!

Topologically, $SO(3)$ is not simply connected while it's double cover $SU(2)$ is. Infact this is a special case of the following general result about Lie groups.

Theorem 1.18. *Given any Lie Group G , there is a simply-connected covering group $\tilde{G}$ with an onto homomorphism,*

$$\rho : \tilde{G} \rightarrow G.$$

Given a projective representation U of G , the the cocycle of projective representation $U \circ \rho$ is always inessential, giving an honest representation of $\tilde{G}$.

The same structure extends to the case of generalized rotations i.e. Lorentz transformations, where $SL(2, \mathbb{C})$ double covers $SO^+(3, 1)$.

1.5. Lie Algebras. Since Lie Groups also have a differentiable structure, we can study their structure by using the tools of calculus. In particular, we have the notion of Lie Algebra of a Lie group. We can obtain this algebra as the tangent space at the identity, which exponentiates to generate the whole Lie group. So there is a two way flow of information.

$$G \begin{array}{c} \xrightarrow{T_I G} \\ \xleftarrow{\exp} \end{array} \mathfrak{g}$$

We will start by defining Lie algebra completely as an algebraic entity with important features that allow the *exponentiation* and other things. We then relate this using differential tools starting from the familiar Lie groups of the last section.

Definition 1.19. A vector space V together with a map $[\cdot, \cdot] : V \times V \rightarrow V$ is called a **Lie algebra**, iff

- a) $[\cdot, \cdot]$ is bilinear.
- b) $[\cdot, \cdot]$ is antisymmetric: $\forall v, w \in V, [v, w] = -[w, v]$.
- c) $[\cdot, \cdot]$ satisfies the **Jacobi identity**:

$$\forall u, v, w \in V : [[u, v], w] + [[v, w], u] + [[w, u], v] = 0.$$

The map $[\cdot, \cdot]$ is called the **Lie bracket**. And we usually denote the Lie algebra by $\mathfrak{g}$.

Example 1.20. The Lie bracket $[\cdot, \cdot] \equiv 0$ makes any vector space into a Lie algebra. This is the **trivial** Lie algebra.

Example 1.21. The matrix vector space $M_n(\mathbb{R})$ has a natural Lie bracket $[A, B] = AB - BA$. The associativity immediately gives out the Jacobi identity,

$$\begin{aligned} [[A, B], C] + [[B, C], A] + [[C, A], B] &= (AB - BA)C - C(AB - BA) \\ &\quad + (BC - CB)A - A(BC - CB) \\ &\quad + (CA - AC)B - B(CA - AC) \\ &= 0. \end{aligned}$$

Example 1.22. The familiar vector space $\mathbb{R}^3$ is also a Lie Algebra with the vector product, $[v, w] = v \times w$.

$$\begin{aligned} [[u, v], w] + [[v, w], u] + [[w, u], v] &= (u \times v) \times w \\ &\quad + (v \times w) \times u + (w \times u) \times v \\ &= 0 \end{aligned}$$

from the usual triple product formula.

Example 1.23. The most theoretical example comes from the space of vector fields on a manifold M : $\mathfrak{X}(M)$. The Lie bracket reads,

$$[v, w](f) = v(w(f)) - w(v(f)), \quad \forall v, w \in \mathfrak{X}(M), f \in C^\infty(M).$$

This translates to how the mixed partial derivatives fail to commute locally!

We can naturally associate Lie algebras to *continuous* groups viz. the Lie groups, especially for matrix groups of the previous section. For this, in the next few examples, we look at the tangent space at identity element of a Lie group G . Recall that, the matrix Lie groups occur mostly as regular submanifolds and their tangent spaces are given by the kernels of their differentials.

Proposition 1.24. *If $F : M \rightarrow N$ is a smooth map, and $c \in N$ is a regular value of F , then $S = F^{-1}(c)$ is a regular submanifold. For $p \in S$,*

$$T_p(S) = \ker(F_* : T_p M \rightarrow T_c N).$$

We will work on the space of tangent vectors of the embedding manifold in the form of velocity vectors of curves on M .

Example 1.25 ($\mathfrak{so}(n)$). Consider the smooth map

$$F : M_n(\mathbb{R}) \rightarrow SM_n(\mathbb{R}), F(A) = \forall A^T$$

. I is in fact a regular value of F and $F^{-1}(I) = O(n)$ is a regular submanifold of $M_n(\mathbb{R})$. With, $T_I M_n(\mathbb{R}) = M_n(\mathbb{R})$ consider the curve,

$$\begin{aligned} \gamma : \mathbb{R} &\rightarrow M_n(\mathbb{R}) \\ t &\mapsto \gamma(t) = I + tA \end{aligned}$$

so that, $\gamma'(0) = A$ is an arbitrary tangent vector of $M_n(\mathbb{R})$ at identity. Now, we want to find the tangent vectors which are kernel to the map,

$$F_* : T_I M_n(\mathbb{R}) \rightarrow T_I SM_n(\mathbb{R}).$$

Note that,

$$\begin{aligned} F(\gamma(t)) &= (I + tA)(I + tA^T) \\ &= I + tA^T + tA + t^2 AA^T \\ F_*(\gamma'(t)) &= (F \circ \gamma)'(t) \\ &= A + A^T. \end{aligned}$$

So,

$$\mathfrak{o}(n) := T_I O(n) = \ker F_* = \{A \in M_n(\mathbb{R}) \mid A = -A^T\},$$

the space of skew-symmetric matrices. This is indeed a vector space with usual matrix product as the Lie bracket. So associated to the Lie group $O(n)$, we found a Lie algebra, written $\mathfrak{o}(n)$. In the same sense, since $O^+(n) = SO(n)$ the Lie algebra associated to $SO(n)$ also works out to be the space of skew-symmetric matrices.

Example 1.26 ($\mathfrak{sl}(n, \mathbb{C})$). Consider the smooth map, $\det : M_n(\mathbb{R}) \rightarrow \mathbb{R}$ as a regular value and $SL(n, \mathbb{R}) = \det^{-1}(1)$. Note that $I \in SL(n, \mathbb{R})$. And,

$$T_I SL(n, \mathbb{R}) = \ker(\det_* : T_I M_n(\mathbb{R}) \rightarrow \mathbb{R})$$

. With $T_I M_n(\mathbb{R}) = M_n(\mathbb{R})$, consider the following smooth curve

$$(1.1) \quad \begin{aligned} \gamma : \mathbb{R} &\rightarrow M_n(\mathbb{R}) \\ \gamma(t) &= I + tA \end{aligned}$$

such that $\gamma(0) = I, \gamma'(0) = A$. So that A is an arbitrary tangent vector in $T_I M_n(\mathbb{R})$. Then,

$$\begin{aligned} \det(\gamma(t)) &= \det(I + tA) \\ &= t^n \chi_{-A} \left(\frac{1}{t} \right) \\ &= 1 + \text{tr}(A) + \cdots + t^n \det(A) \\ \det_*(\gamma(t)) &= \text{tr}(A) \end{aligned}$$

where we have used the form of characterisitic polynomial,

$$\chi_A(t) = |tI - A| = t^n - t^{n-1} \text{tr}(A) + \cdots + \det(A).$$

So, $\mathfrak{sl}(n, \mathbb{R}) := T_I SL(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \text{tr}(A) = 0\}$. This is the Lie algebra associated to $SL(n, \mathbb{R})$ with the usual matrix Lie bracket.

Example 1.27 ($\mathfrak{so}(p, q)$). Let $p + q = n$. Consider now the map,

$$F : M_n(\mathbb{R}) \rightarrow SM_n(\mathbb{R}), F(X) = X\eta X^T.$$

$$\text{Here } \eta = \begin{pmatrix} +1 & & & \\ & +1 & & \\ & & \ddots & \\ & & & -1 \\ & & & & -1 \end{pmatrix} \text{ with signature } (p, q). \eta \text{ is a regular value of}$$

F and $F^{-1}(\eta) = O(p, q)$. Consider the curve,

$$(1.2) \quad \begin{aligned} \gamma : \mathbb{R} &\rightarrow M_n(\mathbb{R}) \\ \gamma(t) &= I + tA \end{aligned}$$

which maps as follows,

$$\begin{aligned} F(\gamma(t)) &= (I + tA)\eta(I + tA^T) \\ F_*(\gamma'(t)) &= A\eta + \eta A^T. \end{aligned}$$

to give $\mathfrak{o}(p, q) := T_I O(p, q) = \{A \in M_n(\mathbb{R}) \mid A\eta + \eta A^T = 0\}$, the Lie algebra associated to $O(p, q)$ and the same extends to $SO(p, q)$.

Example 1.28 ($\mathfrak{su}(n)$). Repeating the same process as in $O(n)$, but with the map,

$$F : M_n(\mathbb{C}) \rightarrow H_n, F(X) = XX^\dagger.$$

H_n is the set of all $n \times n$ hermitian matrices and I is the regular value associated to the submanifold of unitary matrices $U(n)$. We then have,

$$\mathfrak{u}(n) := T_I U(n) = \{T \in M_n(\mathbb{C}) \mid T = -T^\dagger\}$$

the space of skew-hermitian matrices as the Lie algebra associated to $U(n)$. For $SU(n)$ however, the associated Lie algebra turns out to be the set of traceless skew-hermitian matrices.

$\mathfrak{gl}(n, \mathbb{C})$	$\mathfrak{gl}(n, \mathbb{R})$	$\mathfrak{o}(p, q)$ $\mathfrak{so}(p, q)$	$\mathfrak{u}(n)$	$\mathfrak{su}(n)$
$n \times n$ complex matrices	$n \times n$ real matrices	$n \times n$ $X\eta = -\eta X^T$	$n \times n$ skew-hermitian matrices	$n \times n$ traceless skew-hermitian matrices
	$\mathfrak{sl}(n, \mathbb{C})$	$\mathfrak{o}(n)$ $\mathfrak{so}(n)$	$\mathfrak{sl}(n, \mathbb{R})$	
	$n \times n$ traceless complex matrices	$n \times n$ $X = -X^T$	$n \times n$ traceless real matrices	

FIGURE 1. Lie algebras of some common Lie Groups

The tangent space at the identity of a Lie group gave us a simple way to find a Lie algebra. The Lie bracket structure in these cases relied on the matrix nature of vector spaces. But it turns out that we can obtain this Lie bracket structure for any Lie group G not just the matrix ones.

Definition 1.29. Let $(V, [\cdot, \cdot])$ be a Lie algebra. A vector subspace $W \subseteq V$ together with the restriction $[\cdot, \cdot]|_{W \times W}$ is called a **Lie subalgebra** of V iff $\forall w, w' \in W, [w, w'] \in W$.

Definition 1.30. Consider the **left translation**,

$$L_g : G \rightarrow G$$

$$h \mapsto gh.$$

This is a diffeomorphism and allows for the push-forward of vector fields on G ,

$$(L_g)_* : \mathfrak{X}(G) \rightarrow \mathfrak{X}(G).$$

A vector field $X \in \mathfrak{X}(G)$ is called **left-invariant** if $(L_g)_* X = X$ for all $g \in G$.

Proposition 1.31. *For a Lie group G , the space of left-invariant vector fields forms a Lie subalgebra of $\mathfrak{X}(G)$.*

Proof. Let's denote the space of left-invariant vector fields by $\mathfrak{X}_l(G)$. $\mathfrak{X}_l(G)$ is indeed a linear subspace of $\mathfrak{X}(G)$ since $(L_g)_*$ is linear. To show that the Lie algebraic structure is preserved over this subspace, we first claim the following.

Claim 1.32. *Let M be a smooth manifold, and $\phi : M \rightarrow M$ a diffeomorphism. Then for any $X, Y \in \mathfrak{X}(M)$,*

$$\phi_* [X, Y] = [\phi_* X, \phi_* Y].$$

Proof.

$$\begin{aligned} (\phi_* [X, Y])_q (g) &= \phi_{*\phi^{-1}(q)} \left([X, Y]_{\phi^{-1}(q)} \right) (g) \\ &= [X, Y]_{\phi^{-1}(q)} (g \circ \phi) \\ &= X_{\phi^{-1}(q)} (Y(g \circ \phi)) - Y_{\phi^{-1}(q)} (X(g \circ \phi)) \\ [\phi_* X, \phi_* Y]_q (g) &= (\phi_* X)_q (\phi_* Y(g)) - (\phi_* Y)_q (\phi_* X(g)) \\ &= X_{\phi^{-1}(q)} (\phi_* Y(g) \circ \phi) - Y_{\phi^{-1}(q)} (\phi_* X(g) \circ \phi) \\ &= X_{\phi^{-1}(q)} (Y(g \circ \phi)) - Y_{\phi^{-1}(q)} (X(g \circ \phi)) \end{aligned}$$

where in the last step we used a characterization of the facts that $X, \phi_* X$ are ϕ -related and $Y, \phi_* Y$ are ϕ -related! $\square$

Now, if $X, Y \in \mathfrak{X}_l(G)$ then

$$(L_g)_* [X, Y] = [(L_g)_* X, (L_g)_* Y] = [X, Y].$$

Thus, $\mathfrak{X}_l(G)$ is indeed a Lie subalgebra of $\mathfrak{X}(G)$. $\square$

Proposition 1.33. *The space of left-invariant vector fields in a Lie group G is isomorphic to the tangent space $T_e G$, at the identity $e \in G$.*

Proof. For X_e some tangent vector in $T_e G$, define a smooth vector field X such that, $X_g = (L_g)_* X_e$. Then,

$$(L_g)_* X_h = (L_g)_* (L_h)_* X_e = L_{gh} X_e = X_{gh}.$$

Thus, $L_g X = X$, $\implies X \in \mathfrak{X}_l(G)$. This linear surjection is also injective. Because for any $X \in \mathfrak{X}_l(G)$, there is a unique tangent vector $v \in T_e G$, which generates X such that $X_e = v$. This tangent vector is clearly X_e . $\square$

This says that any left-invariant vector fields are determined completely by their value at the identity $e \in G$. And most importantly, for any Lie group G , the two ways to associate a Lie algebra $\mathfrak{g}$ are equivalent (upto isomorphism). We thus define the Lie algebra associated with a Lie group to be either its tangent space at identity, or the space of left-invariant vector fields.

We will now briefly sketch some more ideas which relate Lie algebras and Lie groups.

1.6. Exponentiation. The importance of Lie algebra comes mainly from its infinitesimal data about Lie groups. For example, consider $SO(3)$. It's Lie algebra is spanned by,

$$J_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, J_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, J_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

which are essentially the derivatives at identity, for the rotations along x, y, z axes respectively. The Lie group $SO(3)$ can precisely be generated by the exponentiation,

$$\exp t(n^x J_x + n^y J_y + n^z J_z).$$

Further, it is not necessarily true that $\exp(sJ_x)\exp(tJ_y) = \exp(tJ_y)\exp(sJ_x)$. Infact,

$$\exp(sJ_x)\exp(tJ_y) - \exp(tJ_y)\exp(sJ_x) = st(J_x J_y - J_y J_x) + \mathcal{O}(s^2, t^2).$$

And it is precisely the Lie bracket structure of the Lie algebra that captures the non-commutativity of the Lie group elements.

This exponentiation, thanks to the Lie bracket structure is possible for any Lie algebra. This is done by considering maximal integral curves of left-invariant vector fields. We will only state the result in simpler terms.

Proposition 1.34. *Any Lie group G has an exponentiation map,*

$$\exp : \mathfrak{g} \rightarrow G$$

uniquely determined by the following properties.

- a) $\exp(0)$ is the identity element of G .
- b) $\exp(sx)\exp(tx) = \exp((s+t)x)$ for all $x \in \mathfrak{g}$ and $s, t \in \mathbb{R}$.
- c) $\frac{d}{dt}\exp(tx)|_{t=0} = x$.

Infact, $\exp(\mathfrak{g}) \subseteq G^+$ and for G compact and connected, $\exp(\mathfrak{g}) = G$.

1.7. Lie group to Lie algebra homomorphism.

Definition 1.35. A Lie algebra homomorphism between two Lie groups G and H is a smooth group homomorphism,

$$\rho : G \rightarrow H.$$

This is called a Lie group isomorphism, if there is smooth inverse which is also a group homomorphism.

Example 1.36. The Lie groups $SO(2)$ and $U(1)$ are isomorphic as Lie groups (diffeomorphic to S^1) with the following map,

$$\rho : SO(2) \rightarrow U(1), \begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix} \mapsto e^{i\varphi}.$$

Definition 1.37. If $\mathfrak{g}$ and $\mathfrak{h}$ are Lie algebras, then a linear map $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ is called a Lie algebra homomorphism if $\phi([X, Y]) = [\phi(X), \phi(Y)]$ for all $X, Y \in \mathfrak{g}$. If, in addition, ϕ is one-to-one and onto, then it is called a Lie algebra isomorphism.

Proposition 1.38. If $\rho : G \rightarrow H$ is a Lie group homomorphism, then $\rho_{*e} : \mathfrak{g} \rightarrow \mathfrak{h}$ is a Lie algebra homomorphism.

Note that, this is just a pushforward of the tangent spaces at identity.

Example 1.39. Consider the double cover homomorphism,

$$\rho : SU(2) \rightarrow SO(3).$$

We can compute the map $d\rho : \mathfrak{su}(2) \rightarrow \mathfrak{so}(3)$ explicitly as follows. Let

$$g_t = \exp(-t\sigma_3/2) \in SU(2).$$

$\rho(g_t)$ is defined through its action on the basis of Pauli matrices, $\rho(g_t)\sigma_j = g_t\sigma_jg_t^{-1}$. This can be computed easily to show that,

$$\rho(g_t) = \begin{pmatrix} \cos t & -\sin t & 0 \\ \sin t & \cos t & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

in the basis of Pauli matrices σ_j . This is just the rotation around z axis. Similarly $\exp(-it\sigma_1/2), \exp(-it\sigma_2/2)$ are mapped to rotations around x and y axes. And since any element in $SO(3)$ is a combination of these three rotations, ρ is onto. Now, taking a derivative at identity, i.e. $t = 0$, we have the Lie algebra homomorphism,

$$d\rho : \mathfrak{su}(2) \rightarrow \mathfrak{so}(3), d\rho(-i\sigma_i/2) = J_i$$

which is clearly one-one and onto, showing $\mathfrak{su}(2) \cong \mathfrak{so}(3)$. Note that the Lie bracket structure (commutation rules) of both these algebras are indeed similar to that of quaternions.

There is a generalization to the above features:

- (1) For any Lie algebra $\mathfrak{g}$, there is a unique such Lie group G (upto isomorphism, with Lie algebra $\mathfrak{g}$) that is connected and simply connected.

- (2) All other Lie groups $\tilde{G}$ with $\mathfrak{g}$ as Lie algebra are covered by G through a homomorphism $\rho : G \rightarrow \tilde{G}$ such that $d\rho$ is an isomorphism. For example, take $SU(2)$ and $SO(3)$. The lie algebras, $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ and there is a double cover homomorphism $\rho : SU(2) \rightarrow SO(3)$ where $SU(2)$ is simply connected but $SO(3)$ is not. Indeed $d\rho$ gives the required Lie algebra isomorphism.

1.8. Lie algebra representations. A representation of Lie algebra $\mathfrak{g}$ over a vector space V is defined by the Lie algebra homomorphism,

$$f : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$$

Given a Lie group representation $\rho : G \rightarrow GL(V)$, we can get the corresponding representation of the Lie algebra, through the differential,

$$d\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$$

which as mentioned earlier is indeed a Lie algebra homomorphism. It is also true if a Lie group G is connected and given any Lie algebra representation, we can *exponentiate* this to obtain a representation of G . Lie algebra representations are more simpler to work, since manifolds could be difficult to work with compared to vector spaces. Indeed much of quantum mechanics relies on these structures, and the relation between them.

These Lie groups occurs as *symmetry groups* or as *gauge groups* of the force fields in physics. Electromagnetism and the weak force is unified and described by $SU(2) \times U(1)$, the strong nuclear force by $SU(3)$ and the grand unified efforts to describe the entire standard model (except gravity) by $SU(3) \times SU(2) \times SU(1) \subset SU(5)$. Gravity can also be seen in a sense as a gauge theory over the proper orthochronous lorentz group itself $SO^+(3,1)$ or $SL(2, \mathbb{C})$. Now, we develop the geometric structures, which rely on these Lie groups to describe these forces.

2. GEOMETRY OF BUNDLES

2.1. G-bundle. There is a natural way to build a fiber bundle P using the information of a base manifold M , and Lie group G acting on a typical fiber F .

Definition 2.1. A **principal fiber bundle** or a **G-bundle** is a fiber bundle (P, π, M, F) with a **structure group** G which is a Lie group and a typical fiber F such that:

- (i) G acts freely, and transitively from right on $\pi^{-1}(p) \equiv F_p \cong F$, the fibers of P for all $x \in M$.
- (ii) The *local trivialization* of P , $\psi_U : \pi^{-1}(U) \rightarrow U \times G$ are such that the following diagram commutes:

$$\begin{array}{ccc}
\pi^{-1}(U) & \xrightarrow{\psi_U} & U \times G \\
\uparrow id_{\pi^{-1}U} \times id_G & & \uparrow id_U \times \mu_G \\
\pi^{-1}(U) \times G & \xrightarrow{\psi_U \times id_G} & U \times G \times G
\end{array}$$

i.e. if $\psi_U(u) = (\pi(u), \phi(u))$ then $\pi(ua) = u$,
 $\phi(ua) = \phi(u)a$ for all $u \in \pi^{-1}(U), a \in G$.

We now describe how this definition is a very sophisticated summary of using a Lie group G to construct a fiber bundle.

2.1.1. *Action of G on F .* The third condition amounts to making the right action on F more precise. That is, if $u \in \pi^{-1}(U)$, $\pi(u) = p$ and $\psi_U(u) = (p, g) \in U \times G$, then for any $a \in G$, $ua \in \pi^{-1}(p)$ and $\psi(ua) = (p, ga)$.

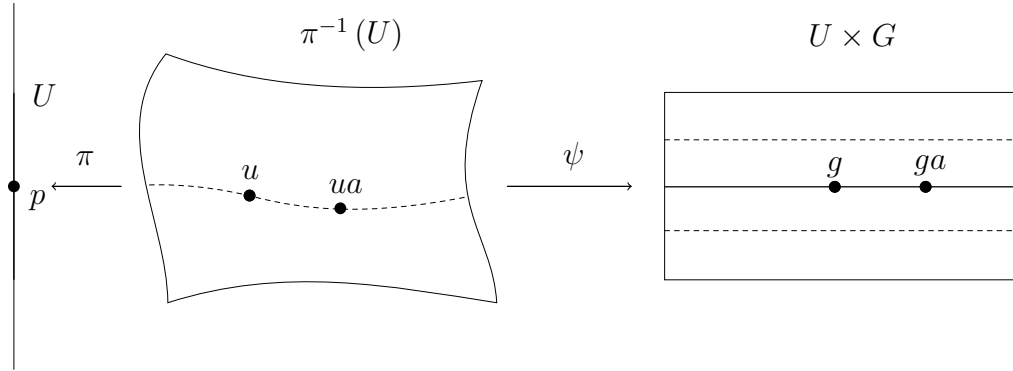


FIGURE 2. G acting on F as a *structure* group

2.1.2. $F_p \cong G$. A free and transitive action essential gives a bijection between G on F . Consider the map,

$$\begin{aligned}
L_g : G &\rightarrow F_p \\
g &\mapsto p \cdot g.
\end{aligned}$$

2.1.3. *Existence of transition functions.* The third condition demands the trivializations to be so that transition functions live on G .

An equivalent definition of G – *bundle* thus follows from the above.

Construction of a G – *bundle* using an open cover of M .

Examples

2.2. **Non Abelian Gauge theory à la Physics.** (Ryder)

2.3. **Connections.**

2.4. **Curvature.**

3. YANG-MILLS THEORY

4. GRAVITY AS A GAUGE THEORY

5. CONCLUSION

Connections to low-dimensional topology. Chern's remarks of natural mathematics and ... hype...